Thyroid Scintigraphy Lec“3”

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Learning Objectives:



By the end of this lecture, the learner should be able to:

- Explain the physiological basis of thyroid radionuclide uptake.
- Describe the physical and biological properties of:

99mTc-pertechnetate
123I-sodium iodide
131I-sodium iodide

- Explain the difference between thyroid trapping and iodine organification.
- Identify appropriate clinical indications for thyroid scintigraphy.
- Prepare a patient correctly before thyroid scintigraphy.
- Perform thyroid imaging using a gamma camera according to an appropriate acquisition protocol.
- Recognize normal thyroid scintigraphic anatomy.
- Identify and characterize hot, warm and cold nodules.
- Recognize scintigraphic patterns of Graves disease, toxic adenoma and toxic multinodular goiter.
- Recognize common artifacts and technical pitfalls.
- Understand the role and limitations of 99mTc compared with radioiodine.
- Perform appropriate radiation-protection procedures.
- Correlate scintigraphic findings with TSH, thyroid function tests and ultrasound.
- Recognize situations in which scintigraphy is inappropriate or should be deferred.
- Produce a technically adequate thyroid scintigraphy report.



The thyroid gland

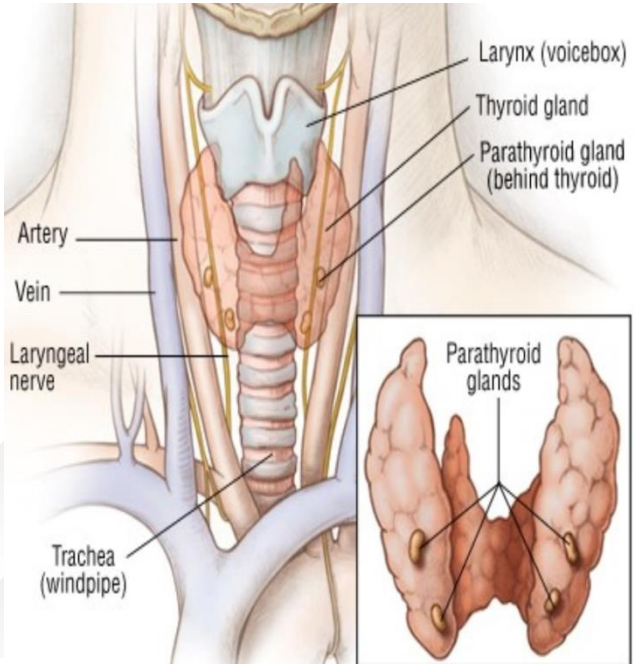
The thyroid gland is located in front of the neck below the thyroid cartilage: The average weight of the thyroid gland, in healthy adults, is 15 gr to 30 gr

It consists of two lobes connected by an isthmus in the middle

The pyramidal lobe, which ascends from the isthmus or the adjacent part of either lobe up to the hyoid bone:

Can be seen in 20 % of the patients

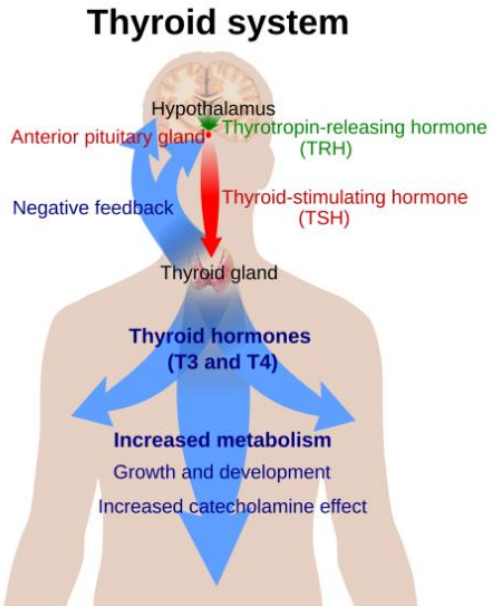
- It consists of two lobes , right and left cone shape lobe: 5x3x2cm.
- Connected by narrow region of gland isthmus 1.2x1.2 cm.
- Situated anterior in the neck at level of C5- T1 vertebra.

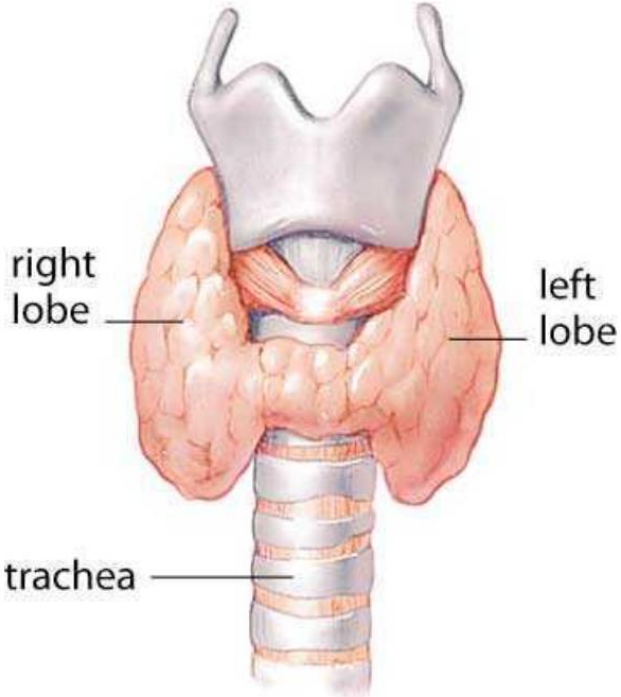


The thyroid is responsible for the production and secretion of thyroid hormones and maintaining iodine homeostasis within the human body.

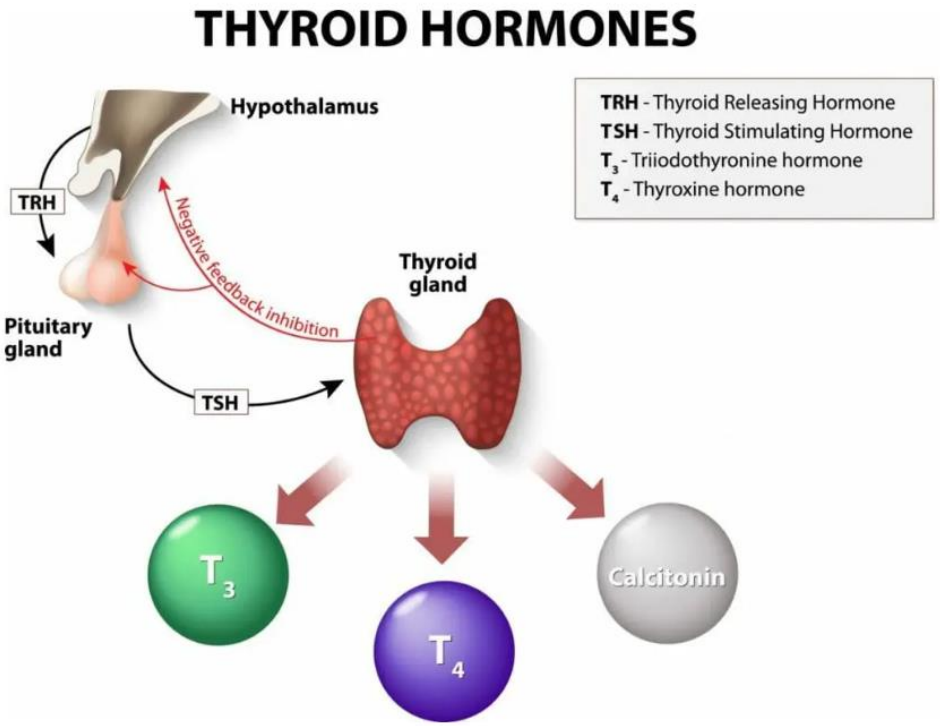
The thyroid produces mainly the inactive thyroid hormone, or thyroxine (T4), and the active thyroid hormone, or triiodothyronine (T3), that regulate human metabolism.

A thyroid disorder occurs when this gland hypersecretes or hyposecretes these hormones





Understanding the physiology is essential.
Iodide enters the thyroid
Iodide in the blood is transported into thyroid follicular cells through the:
Sodium-Iodide Symporter — NIS
NIS
↓
Iodide enters follicular cell
↓
Transport toward colloid
↓
Oxidation
↓
Organification
↓
Thyroglobulin
↓
T3/T4 synthesis



NIS is located on the basolateral membrane of thyroid follicular cells.

TSH stimulates thyroid iodine uptake.



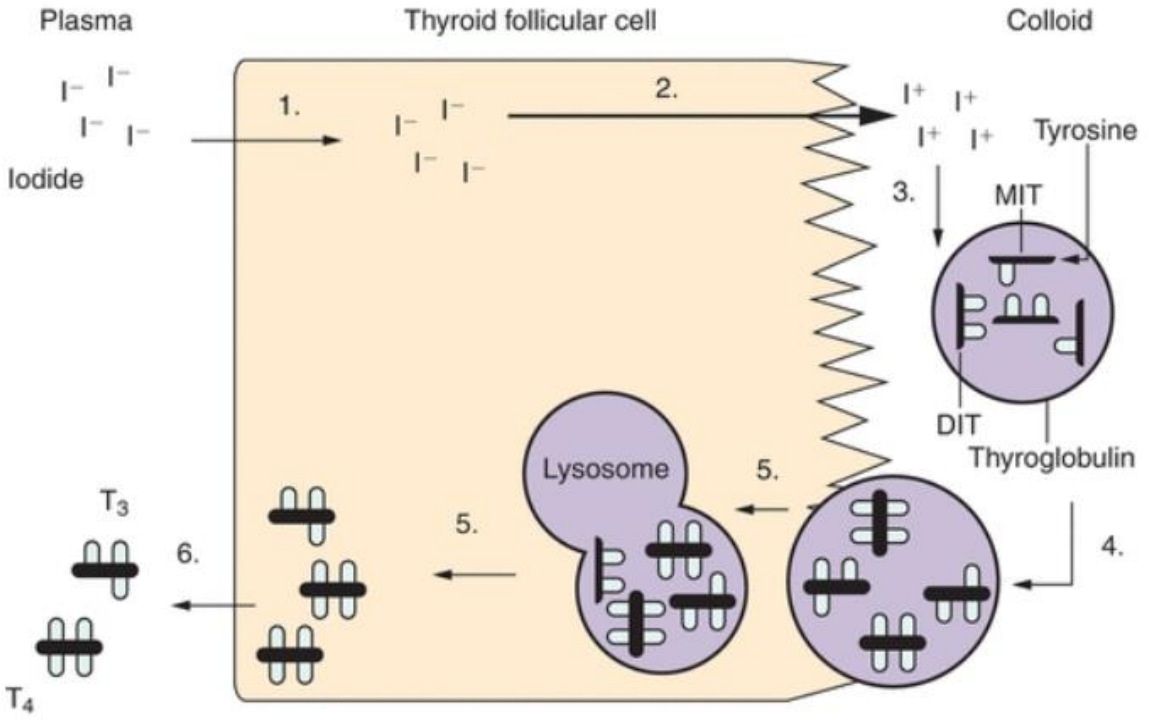

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Typical thyroid function test results for thyroid disorders

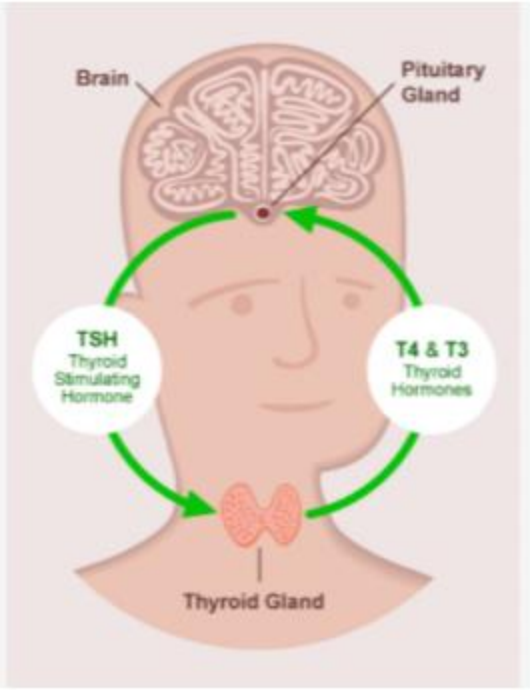
	TSH result	Free T4 result	Free T3 result
Normal thyroid function	Normal range* is 0.45-4.5 mIU/L	Normal range* is 12-30 pmol/L	Normal range* is 2-7 pmol/L
Hyperthyroidism	Low	High	High
Sub-clinical, or mild, hyperthyroidism	Low	Normal	Normal
Hypothyroidism (primary)	High	Low	Low
Hypothyroidism (secondary or tertiary)	Low/normal	Low	Low
Sub-clinical, or mild hypothyroidism	High	Normal	Normal



T4 is the pro-hormone product of the thyroid gland. The thyroid secretes 80 to 100 µg of T4 daily; its half life in the circulation is 6 to 7 days.
Thus missing a morning dose of replacement T4 on the morning of surgery has minimal impact.
T4 is metabolized to biologically active T3 or biologically inactive reverse T3. T3 (3,5,3-triiodothyronine) is formed by extrathyroidal deiodination of T4 (80%) and by direct thyroid secretion (20%).
It is more potent and less protein bound than T4; its half-life in the circulation is 24 to 30 hours. T3 mediates most of the effects of thyroid hormones.



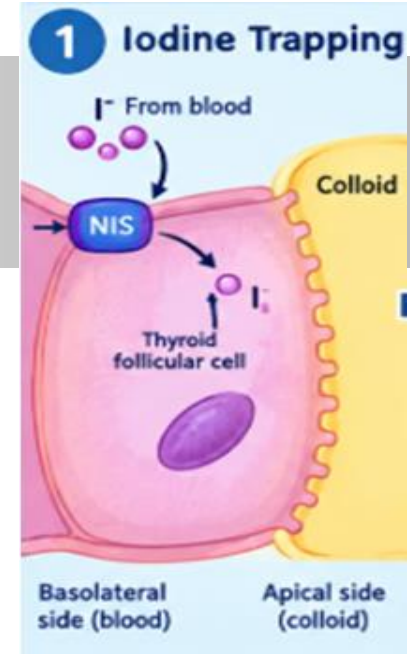
Thyroid hormone synthesis





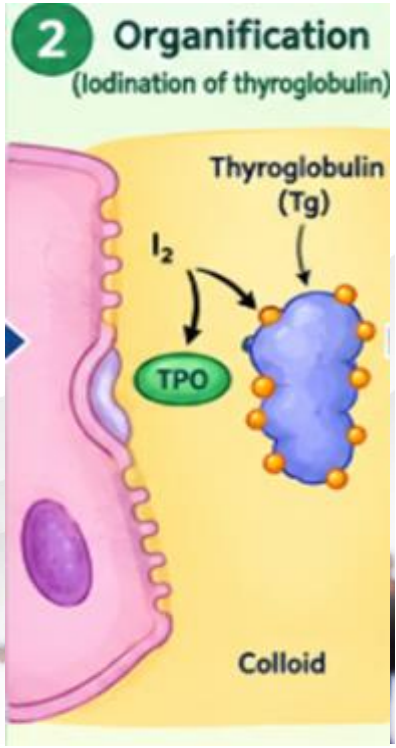
1) Iodine trapping — uptake of iodide

Dietary iodine is absorbed mainly as iodide (I^-) into the circulation. The thyroid actively concentrates iodide from the blood because thyroid hormone synthesis requires iodine.



2) Organification — oxidation and incorporation of iodine

This is the critical biochemical step in which inorganic iodide becomes covalently attached to tyrosine residues in thyroglobulin. Thyroglobulin is a very large glycoprotein synthesized by the follicular cell and secreted into the colloid.It contains many tyrosine residues.These tyrosine residues provide the substrate from which T3 and T4 are eventually produced.

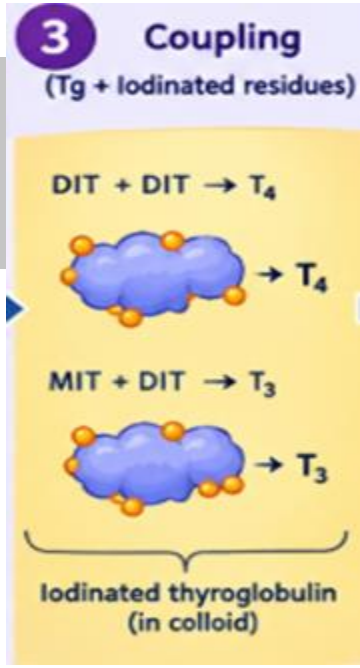


3) Coupling — formation of T3 and T4

The next step is coupling of iodinated tyrosine residues. There are two major reactions.

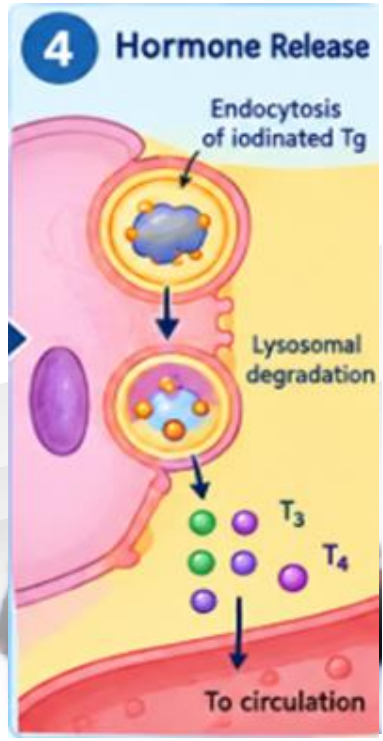
DIT + DIT → T4 it is called:thyroxine / tetraiodothyronine.

MIT + DIT → T3 = triiodothyronine



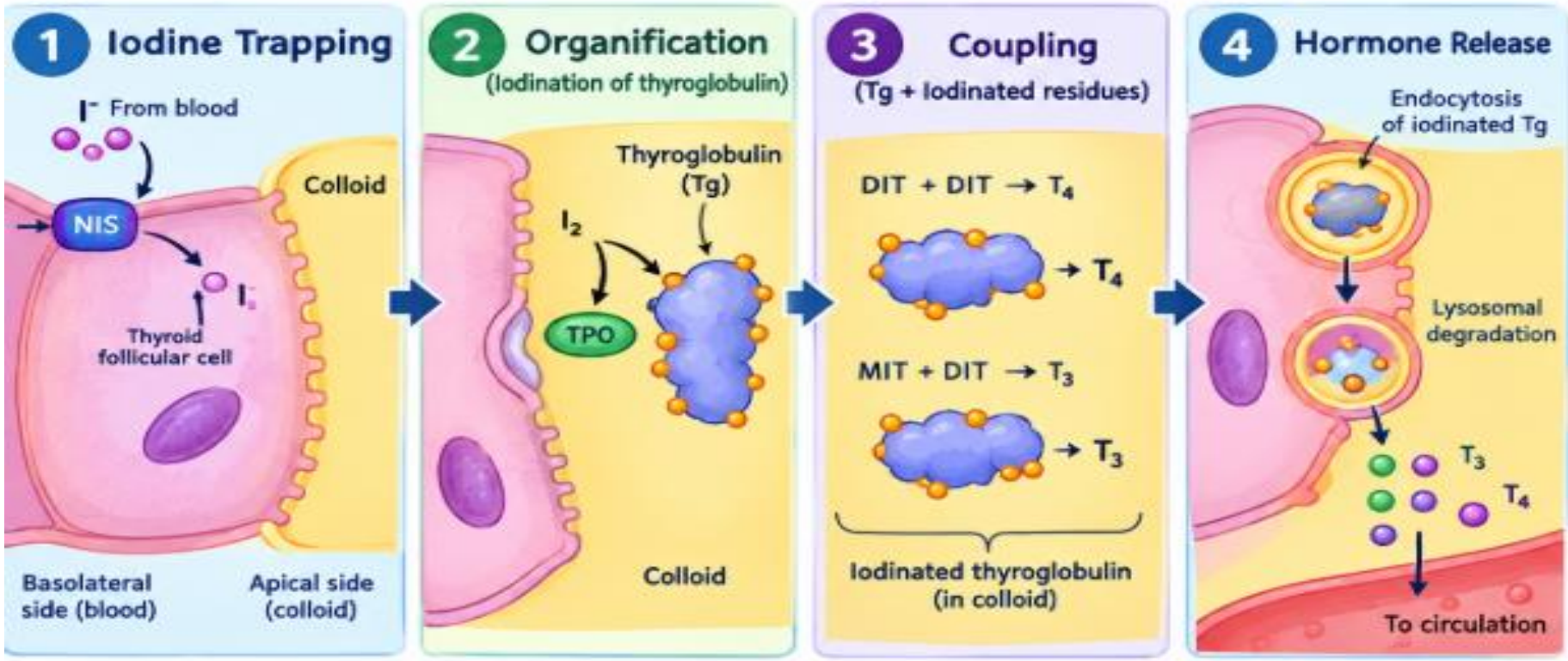
Hormone release

Thyroid hormones T3 and T4 are stored in the colloid bound to thyroglobulin until they are needed. TSH stimulates endocytosis of iodinated thyroglobulin, followed by lysosomal proteolysis that releases T3 and T4 from thyroglobulin.Free T3 and T4 then leave the follicular cell and enter the bloodstream.



Thyroid Hormone Synthesis

Iodine Trapping → Organification → Coupling → Hormone Release



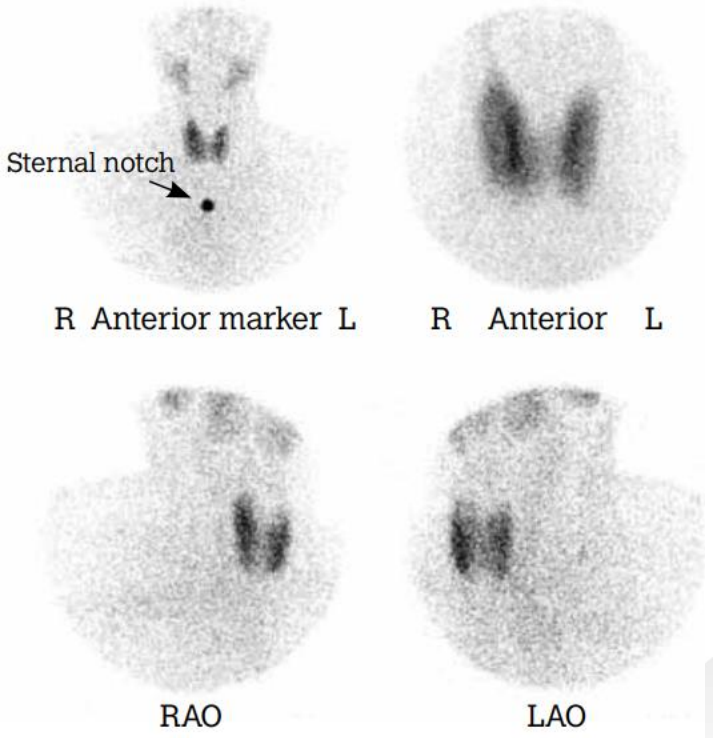
Iodide (I^-) from blood → Trapped by NIS in follicular cell → Organified on Tg (by TPO) → Coupled to form T_3 and T_4 (in colloid) → Endocytosis, lysosomal degradation and release of T_3 and T_4

Iodide (I^-) · Iodine · Thyroglobulin (Tg) · TPO (Thyroid peroxidase) · T_3 · T_4

Thyroid scintigraphy is a functional nuclear medicine imaging study that demonstrates the distribution of radionuclide within thyroid tissue.

It provides information about:

1. Thyroid location
2. Size
3. Shape
4. Functional distribution
5. Focal abnormalities
6. Diffuse abnormalities
7. Ectopic thyroid tissue
8. Autonomous functioning nodules

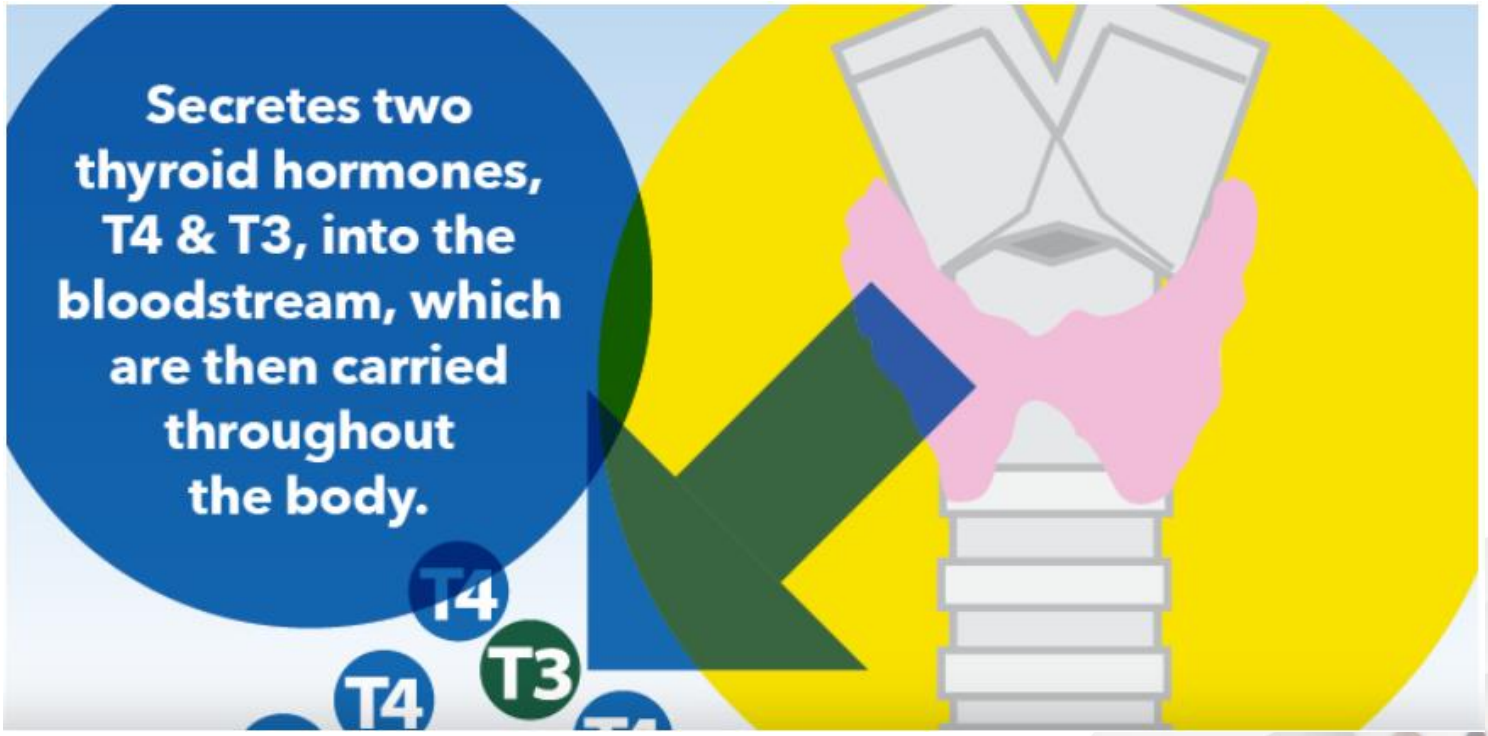


Unlike ultrasound, which primarily demonstrates anatomy, scintigraphy demonstrates function.



The thyroid has a unique ability to concentrate iodide. Radioactive iodine therefore behaves physiologically like stable iodide.

The gamma camera detects photons emitted from the radionuclide and creates a functional map of thyroid tissue.

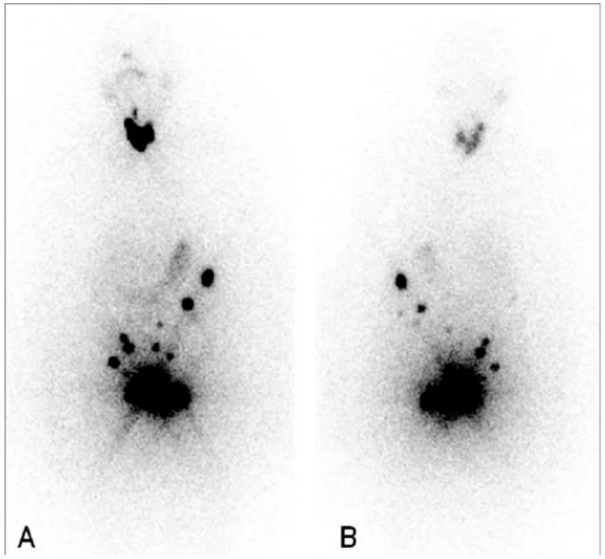


Main Radiopharmaceuticals

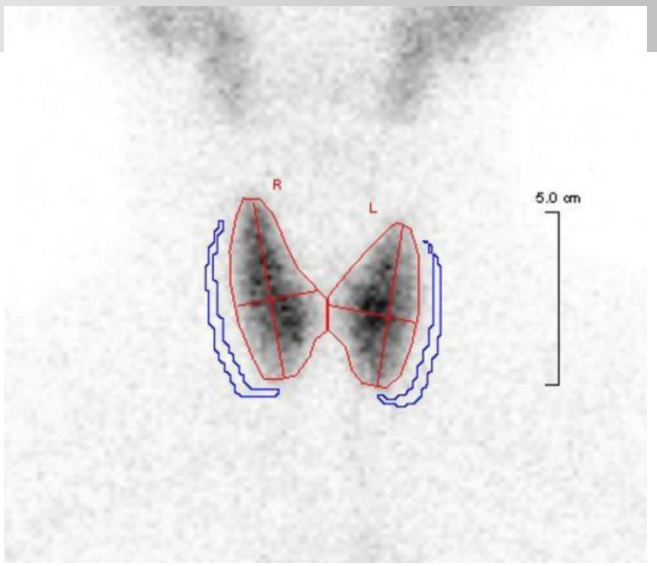
Radiopharmaceutical	Physical half-life	Main photon	Thyroid trapping	Organification
99mTc-pertechnetate	~6 h	140 keV	Yes	No
123I-NaI	~13 h	159 keV	Yes	Yes
131I-NaI	~8 days	364 keV	Yes	Yes



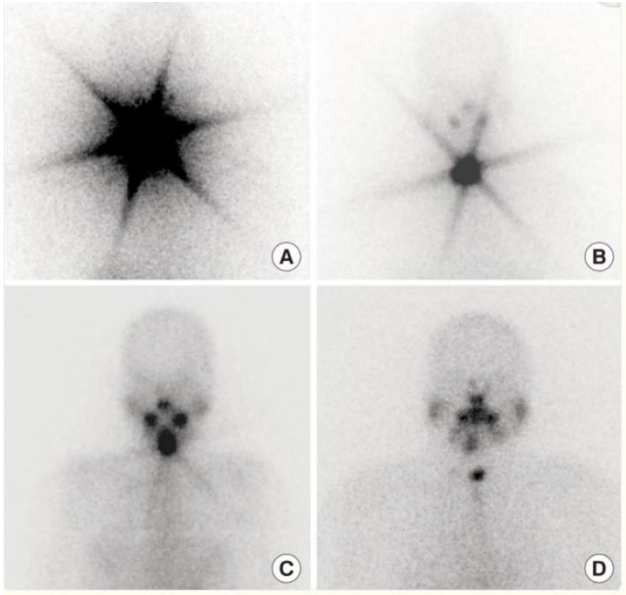
Radioactive iodine-123 (RAI-123) scans. The patient's (A) anterior and (B) posterior RAI-123 imaging displays uptake by thyroid and abdominal masses



Tc99m thyroid scans. anterior imaging.



131I whole body scan images (A–C) with and (D) without star artifact after 1st dose of radioactive iodine treatment.





99mTc-pertechnetate

99mTc-pertechnetate is trapped but NOT organified.
Radioiodine is:
trapped → transported → organified → retained
This difference explains some discordant thyroid scans.

99mTc-Pertechnetate

✓ Mechanism

$99mTcO_4^-$ behaves as an iodide analogue at the level of the Sodium-Iodide Symporter (NIS).
It is actively trapped by thyroid follicular cells but NOT incorporated into thyroglobulin.
Therefore, thyroid retention is relatively short.



99mTc-pertechnetate

Advantages

- ❑ Widely available
- ❑ Relatively inexpensive
- ❑ Short physical half-life
- ❑ Excellent gamma-camera characteristics

140 keV photon is well suited to conventional gamma cameras

- ❑ Low radiation burden compared with diagnostic radioiodine studies
- ❑ Rapid examination



Radioactive iodine
I-123

I-123 sodium iodide is generally the preferred radioiodine imaging agent when available.

Advantages:

- ❑ Excellent gamma energy: 159 keV
- ❑ Physiological iodine handling
- ❑ Allows assessment of iodine trapping and organification
- ❑ Good-quality thyroid imaging
- ❑ Lower radiation burden than I-131 for diagnostic imaging
- ❑ Imaging can generally be performed several hours after administration, with delayed imaging possible depending on the clinical question and protocol.



I-131

I-131 emits:

1) Beta particles
2) Gamma photons

The gamma photon is approximately:

- 364 keV
- Because of its relatively high radiation dose and less favorable imaging characteristics, I-131 is generally not the preferred radionuclide for routine diagnostic thyroid scintigraphy when I-123 or 99mTc is available.

It has major importance in:

- Radioiodine therapy
- Differentiated thyroid cancer
- Selected diagnostic uptake/scan protocols
- Routine diagnostic use should be distinguished from therapeutic use. Current practice parameters emphasize the higher radiation burden of I-131.



99mTc vs I-123

Feature	99mTc-pertechnetate	I-123
Half-life	~6 h	~13 h
Energy	140 keV	159 keV
NIS trapping	Yes	Yes
Organification	**No**	**Yes**
Availability	Excellent	More limited
Cost	Lower	Higher
Imaging	Excellent	Excellent
Examination duration	Short	Longer
Radiation dose	Low	Low
Routine thyroid scan	Very commonly used	Excellent alternative/preferred when available
Organification defects	Cannot evaluate	Can help evaluate

Comparison between 99mTc-pertechnetate and I-123

Radionuclide	Advantages	Disadvantages
Tc-99m pertechnetate	• Less expensive • More readily available • More rapid examination	• Trapped, but not organified • Activity in esophagus or vascular structures can be misleading • Poor image quality when uptake is low
I-123 iodide	• Better for visualization of retrosternal thyroid tissue • Yields better images when uptake is low	• Higher cost • May be less convenient for patient, as delayed imaging at 24 hr is often used • Less readily available • Imaging times are generally longer



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Thank You



